$$|u, b\rangle$$

$$|u, +1\rangle$$

$$|u, -1\rangle$$

$$|d, b\rangle$$

$$|d, +1\rangle$$

$$|d, -1\rangle.$$

The detailed mechanics of what takes place when system meets apparatus may be complicated, but we are free to make some assumptions about how the combined system evolves. Let's assume the apparatus starts in the blank state and the spin starts in the *up* state. After the apparatus interacts with the spin, the final state (by assumption) is

$$|u, +1\rangle.$$

In other words, the interaction leaves the spin unchanged but flips the apparatus to the +1 state. We write this as

$$|u, b\rangle \rightarrow |u, +1\rangle. \qquad (7.28)$$

Similarly, we can require that if the spin is in the *down* state, it flips the apparatus to the −1 state:

$$|d, b\rangle \rightarrow |d, -1\rangle. \qquad (7.29)$$

So by looking at the apparatus after it interacts with the spin, you can tell what the spin was initially. Now, let's assume that the initial spin state is